Name: Nawrin Zawna A N

Host Name:

```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : JARVIS
```

Explain the following concepts in your own word:

1. What is an IP Address?

   An IP (Internet Protocol) address is a unique numerical label assigned to every device connected to a computer network.

2. What is a MAC Address?

   A MAC (Media Access Control) address is a unique, permanent 12-digit physical fingerprint built into your device's network hardware at the factory.

3. What is a Default Gateway?

   A Default Gateway is the node or router that serves as the access point or "exit door" for your local network to connect to outside networks.

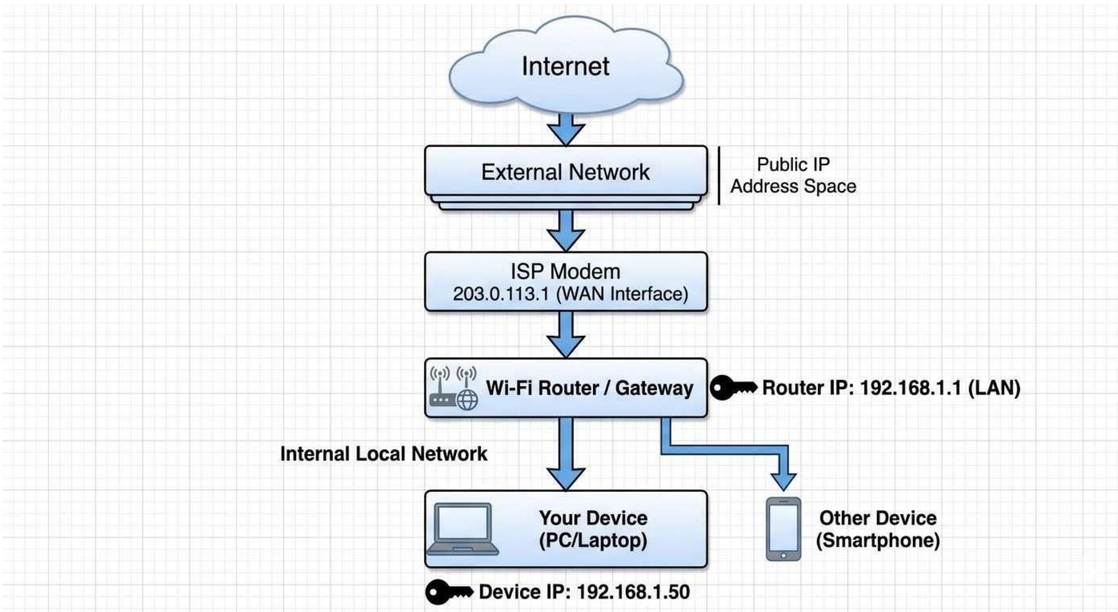
4. What is DNS?

   DNS (Domain Name System) acts as the phonebook of the internet, translating human-readable website names into numerical IP addresses that computers understand.

5. Difference between Public IP and Private IP?

- **Private IP Address:** Used exclusively *inside* your home or local business network to identify local devices like your laptop, phone, and smart TV. It is hidden from the outside world and cannot be accessed directly from the internet.
- **Public IP Address:** A single, unique address assigned to your network router by your internet provider. It represents your entire household or office to the rest of the world, allowing websites and online servers to send data back to your network.

3. Create a simple diagram showing how your device connects to the internet.



Part D: Network Connectivity Test

1. Ipconfig

```
C:\Windows\System32>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

2. Ping google.com

```
C:\Windows\System32>ping google.com

Pinging google.com [2404:6800:4007:80f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:80f::200e: time=27ms
Reply from 2404:6800:4007:80f::200e: time=46ms
Reply from 2404:6800:4007:80f::200e: time=34ms
Reply from 2404:6800:4007:80f::200e: time=39ms
```

3. tracert google.com

```
C:\Windows\System32>tracert google.com

Tracing route to google.com [2404:6800:4007:80f::20
over a maximum of 30 hops:

  1    18 ms     3 ms     2 ms  2401:4900:8839:7c36
  2     *        *        *     Request timed out.
  3    28 ms    50 ms    42 ms  fc00::b2
  4    36 ms    16 ms    24 ms  ^C
C:\Windows\System32>_
```

1. Was the ping successful?

   YES

2. What is the purpose of traceroute?
   The purpose of traceroute is to map the exact path data takes to reach a destination

Readme

It was nice doing this, I got to know where the system configuration are how to take it.
it was useful.

And one more thing, Hope the authorities can understand, like I don’t feel safe to send my system configurations on git or post for it,
So, I HOPE THE COMPANY MEMBER CAN UNDERSTAND THIS.. But I got it how this works